specializes in a single task, multi-agent systems can tackle multiple tasks in parallel. Multiple agents can also provide feedback and evaluation, reducing errors when completing assignments.

As we can see, an AI agent or agent system can be assembled in multiple ways. However, an agent itself can also be assembled using multiple components. In the next section, we'll cover topics ranging from an agent's profile to the actions it may perform, as well as memory and planning.

1.2 Understanding the component systems of an agent

Agents can be complex units composed of multiple component systems. These components are the tools the agent employs to help it complete its goal or assigned tasks and even create new ones. Components may be simple or complex systems, typically split into five categories.

Figure 1.3 describes the major categories of components a single-agent system may incorporate. Each element will have subtypes that can define the component's type, structure, and use. At the core of all agents is the profile and persona; extending from that are the systems and functions that enhance the agent.